# TOPOSCOUT

GNSS SURVEY PLANNING SYSTEM

SITE INTELLIGENCE REPORT

REPORT SG-FVEF4N
Generated July 14, 2026
Data current as of report date

PREPARED FOR

## Outkast Designs

BR

T6N, R4W, Mount Diablo Meridian

| COORDINATES | ELEVATION | STATE PLANE | FEMA ZONE | DECLINATION |
|---|---|---|---|---|
| **38.3459°N 122.3047°W**<br>WGS84 | **66 ft**<br>NAVD88 · MSL | **CA 2 (0402) · N 1,887,923.54 E 6,474,289.05 (US ft, NAD83)** | **Zone X**<br>06055C0510FF | **13.0° E**<br>WMM 2026 |

APN **038050009000** | ZONING **AP** | ACRES **0.481 ac**

BEST GNSS WINDOW **Tue, Jul 14 · 6:00 AM – 10:00 AM** | **PDOP 0.98 · 33 sv** | NEAREST CORS **P261 · 14.1 mi ✓**

REPORT CONTENTS



SCAN FOR KML
Site file · Google Earth
↓ Download KMZ

RECORDER
Napa County Recorder-Clerk · 1127 First St, Napa, CA 94559

NEAREST CORS
P261 · 14.1 mi

GENERATED
July 14, 2026

ASSESSOR
Napa County Assessor · 1127 First St, Napa, CA 94559

NEAREST BM
JT9010 · 0.2 mi

SYSTEM
TopoScout · info@toposcout.com

TOPOSCOUT
★ A SITE INTELLIGENCE REPORT ★

## SITE & SIGNAL REPORT

COORDINATES
**38.3459, -122.3047**
WGS84 / NAD83

ELEVATION
**66 ft**
Above MSL (approx)

MAGNETIC DECLINATION
**13.0° E**
Compass reads 13.0° too far W

FEMA FLOOD ZONE
**X**
AREA OF MINIMAL FLOOD HAZARD · Panel 06055C0510FF

### IONOSPHERIC CONDITIONS

PEAK KP INDEX
**4.7**

AVG KP
**2.3**

SOLAR FLUX F10.7
**138** sfu

TEC (VTEC) ⚠ DEGRADED
**16.8** TECU
-8.6 TECU vs median

ASSESSMENT

**MODERATE ACTIVITY**

Peak Kp 4.7 (avg 2.3). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

### GNSS / DOP SUMMARY

| DATE | AVG PDOP | BEST 4-HR WINDOW | AVG SATS | RATING |
|---|---|---|---|---|
| Tue, Jul 14 | 0.98 | 6:00 AM – 10:00 AM | 33 | Ideal |

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location — orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates.

### 🔥 WILDFIRE PROXIMITY

## No Active Fires Within 150 Miles

No active fires as of report date (July 14, 2026) · Conditions may change — verify before fieldwork

### 🛰 GPS CONSTELLATION HEALTH

## ⚠ 1 Satellite Currently Unhealthy — PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

🌿 Land Cover & Multipath Risk

SKY OBSTRUCTION

MULTIPATH RISK

## Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

## Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

**13°** Minor vegetation — slight horizon reduction

**✓ PDOP Check at 13°: 0.9–0.9 across survey window**

PDOP remains ≤ 0.9 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

| Point | Cover Type | Obstruction | Multipath |
|---|---|---|---|
| Center | Dwarf Scrub | Low | Low |
| North | Dwarf Scrub | Low | Low |
| South | Dwarf Scrub | Low | Low |
| East | Dwarf Scrub | Low | Low |
| West | Dwarf Scrub | Low | Low |

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

## Tuesday, Jul 14

Overcast | 💨 12 mph | GO

High 98°F | Low 56°F | Wind max 12 mph | Gusts 15 mph | Precip 0.00" | Sunrise 5:57 AM | Sunset 8:32 PM

🌘 Waning Crescent 0%

GNSS: Avg PDOP **0.98** · Best 4-hr window 6:00 AM – 10:00 AM (PDOP 0.93, 33 sats avg)

### GNSS / DOP — HOURLY

| | PDOP | SATS | RATING |
|---|---|---|---|
| 6:00 AM | 0.91 | 33 sv | IDEAL |
| 7:00 AM | 1.00 | 32 sv | GOOD |
| 8:00 AM | 0.88 | 35 sv | IDEAL |
| 9:00 AM | 0.93 | 33 sv | IDEAL |
| 10:00 AM | 1.09 | 30 sv | GOOD |
| 11:00 AM | 1.03 | 29 sv | GOOD |
| 12:00 PM | 0.92 | 31 sv | IDEAL |
| 1:00 PM | 1.20 | 24 sv | GOOD |
| 2:00 PM | 0.97 | 32 sv | IDEAL |
| 3:00 PM | 0.93 | 33 sv | IDEAL |
| 4:00 PM | 0.97 | 33 sv | IDEAL |
| 5:00 PM | 0.98 | 34 sv | IDEAL |
| 6:00 PM | 1.03 | 28 sv | GOOD |
| 7:00 PM | 0.92 | 31 sv | IDEAL |

⛅ **GO — GOOD CONDITIONS**

Partly cloudy | Wind: 11.7 mph (gusts 15.4 mph) | Precip: 0" | Temp: 56.2°–97.6°F

⚠️ **FIELD SAFETY WARNINGS**

🌡️ HIGH HEAT 97.6°F — Monitor crew for heat stress

### 🚁 DRONE WIND ANALYSIS

✅ **GO** Peak: 11.7/15.4 mph S

☀ Sunrise: **5:57 AM** ☾ Sunset: **8:32 PM** fly window clipped to daylight

✅ Best flight window: **6:00 AM – 9:00 PM (15 hrs)**

🔄 Wind direction shifting — plan mission headings carefully

| | SUSTAINED / GUSTS mph ⚡=turbulence | DIR |
|---|---|---|
| 5 AM | 1/1 mph | NW |
| 6 AM | 2/3 mph | E |
| 7 AM | 6/7 mph | SE |
| 8 AM | 1/2 mph | SE |
| 9 AM | 2/3 mph | NE |
| 10 AM | 6/6 mph | SSE |
| 11 AM | 5/5 mph | S |
| 12 PM | 9/10 mph | S |
| 1 PM | 9/10 mph | S |
| 2 PM | 9/11 mph | SSW |
| 3 PM | 10/13 mph | SSW |
| 4 PM | 12/15 mph | SW |
| 5 PM | 9/12 mph | SW |
| 6 PM | 3/7 mph | SSW |
| 7 PM | 2/6 mph | SE |
| 8 PM | 8/12 mph | SSE |

BEST WINDOW

## 6:00 AM – 9:00 PM

*Ideal conditions — calm winds, clean operations all day*

AIRSPACE **CLASS A** ▶ FREE TO FLY
Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107
✈ TFR STATUS ✅ **NO TFRs WITHIN 30 MI** Check B4UFLY ↗
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

AI FIELD ANALYSIS — CLAUDE HAIKU 4.5

***GNSS SURVEY FIELD PLANNING REPORT***

*Napa County, California (38.3459, -122.3047)*
*Survey Date: Tuesday, July 14, 2026*
*Report Generated: Tuesday, July 14, 2026*

================================================================================
*VERDICT: GO*
================================================================================

*PRIMARY REASON: Excellent PDOP geometry (0.93–0.98) combined with acceptable ionospheric conditions and favorable weather support survey execution on the planned date.*

================================================================================
***BEST SURVEY WINDOW***
================================================================================

*TIME RANGE: 06:00 AM – 10:00 AM (Pacific Daylight Time)*

*RATIONALE: PDOP reaches optimal 0.93 during this 4-hour window with 33 satellites in view across GPS, GLONASS, Galileo, and BeiDou constellations. Morning window occurs before peak heat and wind; weather conditions (overcast, 12 mph max wind) are stable and pose no operational constraint.*

================================================================================
*BEST 4-HOUR STATIC GNSS WINDOW*
================================================================================

*WINDOW: 07:00 AM – 11:00 AM PDT*

***OCCUPATION DETAILS:***
*• PDOP range: 0.93–0.96 (excellent)*
*• Satellite count: 32–33*
*• Recommended session length: 2 hours minimum per station (extended to 2.5 hours recommended due to moderate geomagnetic activity Kp 4.7 and TEC -8.6 TECU anomaly to mitigate ionospheric delay bias)*
*• Elevation mask: 13° (terrain-optimized for dwarf scrub land cover with low multipath risk)*
*• Note: PRN 11 unhealthy and auto-excluded by receiver; PDOP already reflects reduced constellation. PRN 3 maintenance windows (past and future) do not impact July 14 operations.*

================================================================================
***DRONE OPERATIONS ASSESSMENT***
================================================================================

*STATUS: ACCEPTABLE*

*WEATHER: Overcast skies at 98°F high / 56°F low with 12 mph max wind — well within typical UAS operating limits. No precipitation forecast. Visibility adequate for visual line-of-sight operations.*

*AIRSPACE: Confirm airspace clearance and notify local Napa County authorities per Part 107 / local procedures. No wildfire smoke or air quality constraints (AQI 50, Good).*

================================================================================
***FIELD SAFETY WARNINGS***
================================================================================

*• Heat stress risk: Peak temperature 98°F. Provide crew with adequate hydration, shade, and rest cycles during mid-day setup/breakdown.*
*• Moderate geomagnetic activity (Kp 4.7): Position solutions may exhibit transient degradation; validate RTK/static fixes and repeat occupations if post-processing shows anomalies. Do not rely on single-epoch ambiguity resolution during high Kp periods.*

• *Equipment check: Verify all receiver antennas are functioning and aligned to clear sky view (13° mask verified). PRN 11 will be auto-excluded; ensure firmware is current.*

================================================================================

***SUMMARY***

================================================================================

*Conditions on Tuesday, July 14, 2026, are favorable for GNSS survey operations in Napa County. Excellent geometry (PDOP 0.93–0.98) and stable weather support a GO verdict. Execute primary static sessions between 07:00 AM and 11:00 AM PDT, extending occupation times to 2.5 hours per station to compensate for moderate ionospheric activity. Terrain-based 13° elevation mask and low multipath environment at the dwarf scrub site support high-quality observations. Monitor crew for heat stress and validate solutions in post-processing given the Kp 4.7 activity. No wildfire impacts or air quality constraints detected. Proceed with survey as planned.*

## COUNTY RECORDING OFFICES

RECORDER / REGISTER OF DEEDS

Napa County Recorder-Clerk
1127 First St, Napa, CA 94559

ASSESSOR / TAX OFFICE

Napa County Assessor
1127 First St, Napa, CA 94559

## ⚡ Nearby Control Points · 38.3459, -122.3047

### CORS STATIONS — WITHIN 30 MILES

| QR / LINK | STATION ID | NAME / LOCATION | DISTANCE |
|---|---|---|---|
| NGS ↗ | **P261** | HUNTERHILLCN2004 | 14.1 mi |
| NGS ↗ | **P198** | PETALUMAIRCN2004 | 17.5 mi |
| NGS ↗ | **OHLN** | OHLONE PARK | 23.5 mi |
| NGS ↗ | **P196** | MEACHUMLFLCN2006 | 24.0 mi |
| NGS ↗ | **CASR** | SANTA ROSA CA | 24.9 mi |
| NGS ↗ | **P267** | DIXONAVIATCN2005 | 26.3 mi |

### NGS BENCHMARKS — WITHIN 10 MILES

| QR / LINK | PID | DESIGNATION | ELEVATION | ORDER | DISTANCE |
|---|---|---|---|---|---|
| Datasheet ↗ | **JT9010** | "13" | 19.100m / 62.7ft NAVD88 | Order | 0.2 mi |
| Datasheet ↗ | **JT9018** | "13 E" | 17.970m / 59.0ft NAVD88 | Order | 0.2 mi |
| Datasheet ↗ | **JT9009** | "13" | 17.900m / 58.7ft NAVD88 | Order | 0.2 mi |
| Datasheet ↗ | **JT9017** | "13 D" | 17.890m / 58.7ft NAVD88 | Order | 0.2 mi |
| Datasheet ↗ | **JT9034** | "20 A" | 20.730m / 68.0ft NAVD88 | Order | 0.3 mi |
| Datasheet ↗ | **JT9012** | "13" | 19.060m / 62.5ft NAVD88 | Order | 0.4 mi |
| Datasheet ↗ | **JT9043** | "25 A" | 18.710m / 61.4ft NAVD88 | Order | 0.4 mi |
| | **JT9016** | "13 D" | 18.200m / 59.7ft NAVD88 | Order | 0.5 mi |

| QR / LINK | PID | DESIGNATION | ELEVATION | ORDER | DISTANCE |
|---|---|---|---|---|---|
| Datasheet ↗ | | | | | |

## FEMA FLOOD ZONE

**X — AREA OF MINIMAL FLOOD HAZARD**

Panel: **06055C0510FF**
Effective: Sep 29, 2010
View FEMA Map ↗

## DATA COLLECTOR IMPORT — NGS BENCHMARKS

↓ Download .txt File

```
────────────────────────────────────────────────────────────────────────
NGS BENCHMARK CONTROL POINTS — Copy / Paste to Data Collector
Datum: NAD83  |  Vertical: NAVD88  |  Source: NGS NDE API
────────────────────────────────────────────────────────────────────────
PID      NAME                    LAT           LON           ELEV_FT  ORDER
────────────────────────────────────────────────────────────────────────
JT9010   13                        38.348333    -122.302778      62.7
JT9018   13 E                      38.348333    -122.302778      59.0
JT9009   13                        38.346667    -122.301111      58.7
JT9017   13 D                      38.345000    -122.301111      58.7
JT9034   20 A                      38.345000    -122.309444      68.0
JT9012   13                        38.351667    -122.304444      62.5
JT9043   25 A                      38.340000    -122.302778      61.4
JT9016   13 D                      38.341667    -122.297778      59.7
────────────────────────────────────────────────────────────────────────
```

## USGS LAKES & RESERVOIRS — within 150 miles

| STATION | DIST | LEVEL | FULL POOL | LINK |
|---|---|---|---|---|
| **CLEAR LK A LAKEPORT CA** #11450000 | 58.1 mi | 4.98 ft | N/R | USGS ↗ |

## CALIFORNIA RESERVOIRS (DWR/CDEC) — within 150 miles

| RESERVOIR | DIST | ELEVATION | FULL POOL | STORAGE | % CAP |
|---|---|---|---|---|---|
| **Lake Del Valle** | 60.6 mi | 696.82 ft (-9 ft) | 706 ft | 36K af | **67%** |
| **Folsom Lake** | 66.5 mi | 447.31 ft (-19 ft) | 466 ft | 778K af | **80%** |
| **Lake Oroville** | 93.4 mi | 859.42 ft (-41 ft) | 900 ft | 2,834K af | **80%** |
| **New Melones Lake** | 100.1 mi | 1,019.74 ft (-68 ft) | 1,088 ft | 1,659K af | **69%** |
| **Black Butte Lake** | 101.2 mi | — | 228 ft | 73K af | **54%** |

## ARMY CORPS RESERVOIRS (USACE) — within 150 miles

| RESERVOIR | DIST | POOL ELEV | FULL POOL | STORAGE | % CAP |
|---|---|---|---|---|---|
| **Del Valle** | 64 mi | 696.8 ft | 695.01 ft | — | **100%** |
| **Camanche** | 89 mi | 230.2 ft | 221.86 ft | — | **104%** |
| **Englebright Lake-Pool** | 95 mi | 525.1 ft | 522.35 ft | — | **101%** |
| **Farmington Dam-Pool** | 99 mi | 124.4 ft | 117.09 ft | — | **106%** |
| **Black Butte-Pool** | 101 mi | 456.7 ft | 458.36 ft | — | **100%** |
| **Don Pedro** | 137 mi | — | N/R | — | **—** |

## USGS GROUNDWATER WELLS — within 15 miles

| STATION | DIST | WELL DEPTH | WATER LEVEL | DATE | AQUIFER |
|---|---|---|---|---|---|
| **005N004W32A002M** | 7 mi | 200 ft | 87.71 ft BGS | 2014-11-05 | N100CACSTL |
| **005N004W28R001M** | 7 mi | 51 ft | 41.90 ft BGS | 1977-10-03 | N100CACSTL |
| **004N004W04C001M** | 8 mi | 80 ft | 11.20 ft BGS | 1977-10-03 | N100CACSTL |
| **004N004W05B001M** | 8 mi | — | 54.10 ft BGS | 1977-10-03 | N100CACSTL |
| **005N004W31R001M** | 8.1 mi | 295 ft | — | — | — |
| **004N004W05D002M** | 8.3 mi | 60 ft | 14.60 ft BGS | 1977-10-03 | N100CACSTL |
| **004N004W06B001M** | 8.3 mi | 201 ft | 21.86 ft BGS | 2022-06-07 | — |
| **004N004W02L001M** | 8.8 mi | — | 9.90 ft BGS | 1977-10-02 | N100CACSTL |

### ⬡ AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

*The Napa Valley Cachuma–Salinas Limestone (N100CACSTL) is a Tertiary marine carbonate formation that underlies portions of the Coast Ranges and Napa Valley. It formed in shallow seas during the Eocene–Oligocene and has been uplifted and fractured by tectonic activity, creating localized zones of moderate permeability. Water availability at this site is moderate but variable, with significant depth-to-water fluctuations between 1977 and 2022 indicating response to regional groundwater demand and climate cycles.*

***FORMATION***

| | |
|---|---|
| ***Name*** | *Napa Valley Cachuma–Salinas Limestone* |
| ***Age / Rock*** | *Eocene–Oligocene marine limestone and calcareous shale; Tertiary age.* |
| ***Extent*** | *150–300 sq mi (est. USGS/DWR)* |

***DEPTH & RECHARGE***

| | |
|---|---|
| ***Depth Range*** | *51–295 ft (5–29 floors)* |
| ***Typical Depth*** | *100–150 ft (10–15 floors)* |
| ***Recharge Source*** | *Precipitation on outcrop areas + lateral flow from upland carbonate and siliciclastic sources* |
| ***Transit Time*** | *20–50 years to 100 ft depth (est. USGS/DWR)* |

***WATER QUALITY***

| | |
|---|---|
| ***pH*** | *est. 7.2–7.8 (est. USGS/DWR)* |

| | |
|---|---|
| ***Hardness*** | *est. 250–450 mg/L $CaCO_3$ (est. USGS/DWR)* |
| ***TDS*** | *est. 350–650 mg/L (est. USGS/DWR)* |
| ***Notes*** | *No sample data — typical for formation. Carbonate aquifers often yield moderately hard, neutral to slightly alkaline water.* |

***TREND***

| | |
|---|---|
| ***Trend*** | *Water levels show long-term decline from 1977 (9.90–54.10 ft BGS) to 2022 (21.86–87.71 ft BGS), indicating sustained groundwater depletion of ~30–45 ft over 45 years.* |
| ***Source*** | *USGS NWIS well records and local monitoring data* |
| ***Seasonal Swing*** | *10–20 ft intra-annual swing (est. USGS/DWR regional)* |

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

## LAND SUBSIDENCE & AQUIFER COMPACTION

**✓ LOW SUBSIDENCE RISK — No known critically overdrafted basin**

Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

**⬡ AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU**

*The North Coast Coastal Ranges aquifer system (N100CACSTL) in this region is characterized by well-drained soils and moderate recharge conditions typical of the California North Coast hydrologic region. The site's location outside any critically overdrafted groundwater basin indicates that historical groundwater extraction has not exceeded sustainable yield, preventing the clay layer compression and permanent aquifer storage loss that drives subsidence in over-pumped basins. For surveyors operating in this area, this means published elevation benchmarks are expected to remain stable and reliable for survey control.*

***SUBSIDENCE ASSESSMENT***

| | |
|---|---|
| ***Susceptibility*** | *Low* |
| ***Primary Mechanism*** | *None - consolidated terrain* |
| ***Annual Rate*** | *<1 mm/yr (est. USGS)* |
| ***Historic Loss*** | *None documented* |
| ***Aquifer Compaction*** | *None - aquifer not over-pumped* |
| ***SGMA Status*** | *Not applicable - not in SGMA basin* |

***BENCHMARK IMPLICATIONS***

| | |
|---|---|
| ***Monument Risk*** | *Low - monuments stable* |
| ***Elevation Validity*** | *Good - no active subsidence* |
| ***GNSS Recommendation*** | *Standard benchmark verification adequate* |

***SURVEY ACTION***

| | |
|---|---|
| ***Required Action*** | *No special action required* |
| ***Dataset*** | *CA DWR SGMA Bulletin 118 2024.1* |
| ***Coverage*** | *California critically overdrafted basins only - full US coverage pending* |

## STRUCTURAL & ENGINEERING PARAMETERS

**SEISMIC (ASCE 7-22)**

| | |
|---|---|
| **Site Class** | D (default) |
| **Risk Category** | II |

**WIND / SNOW / ICE / FROST (ASCE 7-22)**

| | |
|---|---|
| **Basic Wind Speed** | 92 mph |
| **Ground Snow Load** | 5.8 psf |
| **Winter Wind (W2)** | 0.35 |
| **Ice Thickness** | None |
| **Frost Depth** | **12 in minimum** |

Est. from IECC Zone 3C — AFI data unavailable
Verify with local building official before construction

**Source: ASCE 7-22 Hazard Tool · gis.asce.org**

**CLIMATE ZONE (IECC 2021):** Zone 3C · Marine · Marine - mild year-round, coastal influence (SF, LA coast) County: Napa

**⬡ AI CLIMATE SUMMARY — RESIDENTIAL DESIGN**

This marine climate has mild winters and summers with consistent coastal breezes, so your primary envelope concern is managing solar gain and moisture rather than extreme temperature swings—prioritize high-performance windows on south and west facades while minimizing glass on east and west exposures where afternoon sun can overheat living spaces. The moderate but notable 92 mph wind load means your envelope detailing around windows, doors, and roof connections needs to be robust, and any projections like balconies should be designed with wind resistance in mind. Your best passive design move is cross-ventilation: design buildings with operable windows on opposing facades to capture coastal breezes for natural cooling through most of the year, which will dramatically reduce mechanical cooling needs in this temperate climate.

**⬡ AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU**

*This coastal marine site presents well-drained gravelly loam soils anticipated to support conventional shallow foundations, though seismic parameters are currently unavailable and may warrant confirmation with the local building official. Moderate wind exposure and low snow loads suggest standard structural design approaches, pending geotechnical investigation to confirm site class and bearing capacity.*

***FOUNDATION***

| | |
|---|---|
| ***Type*** | *Conventional spread footing anticipated* |
| ***Basis*** | *Well-drained Coombs gravelly loam with 2–5% slopes suggested to support shallow foundations; however, seismic design category is null and may warrant consultation with a licensed engineer to confirm appropriate foundation detailing and embedment depth (estimated frost line: 12 in).* |

***STRUCTURAL IMPLICATIONS***

| | |
|---|---|
| ***Seismic*** | *Seismic design parameters (SDC, Ss, S1, SDS, SD1, PGA) are null for this location; recommend confirming design earthquake parameters with the local building official or USGS before final structural design. Standard detailing may be adequate, pending confirmation.* |
| ***Wind Exposure*** | *Exposure C (open terrain with scattered obstructions) anticipated per ASCE 7-22; verify site topography and surrounding terrain to confirm exposure classification.* |
| ***Wind*** | *92 mph design wind speed (3-second gust, Risk Category II) anticipated—standard wind load design likely adequate. No hurricane exposure anticipated for this inland location.* |
| ***Snow*** | *Ground snow load of 5.8 psf anticipated with winter wind W2 = 0.35; standard roof design and drainage likely adequate for this low-to-moderate load. Consult engineer for specific roof configuration.* |

***INVESTIGATIONS & INSPECTIONS***

| | |
|---|---|
| ***Geotech Required*** | *Yes—geotechnical investigation recommended to confirm soil bearing capacity, site class designation, and foundation design parameters, particularly given null seismic parameters.* |
| ***Liquefaction Screen*** | *Liquefaction screening recommended given Site Class D assignment and coastal location; shallow groundwater and saturated conditions may warrant detailed evaluation. Consult a licensed geotechnical engineer.* |
| ***Special Inspections*** | *May be required depending on final seismic design category determination; recommend confirming with local building official and structural engineer once seismic parameters are established.* |
| ***Code Edition*** | *ASCE 7-22 / IBC 2021 (planning reference only)* |
| ***Site Class Note*** | *Site Class D assumed for planning—geotechnical investigation strongly recommended to confirm soil profile, classification, and suitability for proposed use.* |

## SOILS — USDA SOIL SURVEY

**MAP UNIT:** Coombs gravelly loam, 2 to 5 percent slopes

**SERIES:** Coombs

**DRAINAGE CLASS:** Well drained

**HYDRO GROUP:** C

**SOIL ORDER:** Alfisols

**SUBORDER:** Xeralfs

### ⬡ AI SOIL INTELLIGENCE — CLAUDE HAIKU

*Coombs gravelly loam is a well-drained Alfisol (Xeralf) formed in alluvial and colluvial material derived from mixed sources, commonly found on gentle valley and foothill slopes in Mediterranean climates. This soil features a gravelly loam surface layer over an argillic (clay-enriched) substratum, providing moderate bearing capacity suitable for light to moderate structural loads. Surveyors and developers should expect firm, stable ground with good drainage but should account for gravel content in excavation planning and verify subsurface clay behavior under moisture conditions.*

*Mixed alluvial and colluvial parent material on 2–5% slopes; typically supporting oak woodland and grassland vegetation.*

***DRAINAGE***

| | |
|---|---|
| ***Class*** | *Well drained* |
| ***Seasonal Saturation*** | *None within 60 inches* |
| ***Flooding Risk*** | *Minimal on 2–5% slopes* |
| ***Ponding Risk*** | *None* |

***ENGINEERING***

| | |
|---|---|
| ***Bearing Capacity*** | *2,500–3,500 psf (shallow foundations)* |
| ***Rating*** | *Good* |
| ***Shrink-Swell*** | *Low to Moderate — argillic substratum present* |
| ***Frost Action*** | *Low* |
| ***Cut Suitability*** | *Good (gravel-rich surface layer)* |
| ***Fill Suitability*** | *Fair (argillic substratum — moisture control required)* |

***SEPTIC***

| | |
|---|---|
| ***Suitability*** | *Conditional* |
| ***Notes*** | *Perc test required — clay substratum may limit percolation; depth to argillic layer and seasonal moisture regime critical for design* |

***HAZARDS***

| | |
|---|---|
| ***Excavation*** | *Moderate — 20–35% gravel content in upper profile* |
| ***Expansive Clay*** | *None anticipated* |
| ***Erosion*** | *Low on 2–5% slopes; moderate risk in disturbed or denuded areas* |
| ***Other*** | *None* |

Data sources: USGS National Water Information System · waterdata.usgs.gov · California DWR / CDEC · cdec.water.ca.gov · USACE CWMS · water.usace.army.mil · USGS NWIS Groundwater · waterservices.usgs.gov · USDA Web Soil Survey · sdmdataaccess.nrcs.usda.gov · CA DWR SGMA Bulletin 118 · gis.water.ca.gov · ASCE 7-22 Hazard Tool · gis.asce.org · USGS Design Maps · earthquake.usgs.gov · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision
Water levels current as of report run: Jul 14, 2026, 7:11 PM UTC
* Live pool elevation not reported by this reservoir — full pool design elevation shown

## 🛢 OIL & GAS WELLS California — within 10mi radius

| 8 | 0 | 7 |
|---|---|---|
| total | active | plugged |

By status: **7** Plugged **1** Idle

By type: 8 DH

| Lease / Well | Status | Type | Operator | Depth | Dist |
|---|---|---|---|---|---|
| Mobil Oil Corporation | **Plugged** | DH | Mobil Oil Corporation | — | 7.2 mi |
| Mobil Oil Corporation | **Plugged** | DH | Mobil Oil Corporation | — | 7.2 mi |
| McKenzie-Hazard Syndicate | **Plugged** | DH | McKenzie-Hazard Syndicate | — | 7.4 mi |
| Mobil Oil Corporation | **Plugged** | DH | Mobil Oil Corporation | — | 8.6 mi |
| Mobil Oil Corporation | **Plugged** | DH | Mobil Oil Corporation | — | 8.8 mi |
| Mobil Oil Corporation | **Plugged** | DH | Mobil Oil Corporation | — | 9.5 mi |
| Capell Oil Corp. | **Idle** | DH | Capell Oil Corp. | — | 9.7 mi |
| Cordelia Oil & Gas Co. | **Plugged** | DH | Cordelia Oil & Gas Co. | — | 12 mi |

### AI Assessment — Well Proximity

Well proximity is not a planning concern for this site. The nearest well is a plugged Mobil Oil Corporation dry hole 7.2 miles away, which is well beyond any actionable distance threshold. With no active producers, injection wells, or idle wells within relevant review distances, you should focus site planning attention on other factors.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

### ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

| 6 | 6 | 12 |
|---|---|---|
| Endangered | Threatened | Total w/ Critical Habitat |

6 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

| COMMON NAME | *SCIENTIFIC NAME* | STATUS |
|---|---|---|
| California tiger Salamander | *Ambystoma californiense* | **Endangered** |
| Conservancy fairy shrimp | *Branchinecta conservatio* | **Endangered** |
| Contra Costa goldfields | *Lasthenia conjugens* | **Endangered** |
| Soft bird's-beak | *Cordylanthus mollis ssp. mollis* | **Endangered** |
| Suisun thistle | *Cirsium hydrophilum var. hydrophilum* | **Endangered** |
| Vernal pool tadpole shrimp | *Lepidurus packardi* | **Endangered** |
| Alameda whipsnake (=striped racer) | *Masticophis lateralis euryxanthus* | **Threatened** |
| California red-legged frog | *Rana draytonii* | **Threatened** |
| Delta smelt | *Hypomesus transpacificus* | **Threatened** |
| Northern spotted owl | *Strix occidentalis caurina* | **Threatened** |
| Vernal pool fairy shrimp | *Branchinecta lynchi* | **Threatened** |
| western snowy plover | *Charadrius nivosus nivosus* | **Threatened** |

#### AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Planning Assessment **Section 7 Consultation Triggers:** If the proposed project involves federal permitting, funding, or licensing, formal Section 7 consultation with USFWS will be required given the presence of six listed endangered species and six threatened species with designated or proposed critical habitat within 25 miles of the site. **Most Planning-Relevant Species:** The vernal pool complex-dependent species (California tiger salamander, Conservancy fairy shrimp, vernal pool tadpole shrimp, and vernal pool fairy shrimp) and aquatic species (Delta smelt, California red-legged frog) warrant immediate screening, as these are most sensitive to hydrology, water quality, and land-use changes typical in development scenarios. Alameda whipsnake should also be prioritized given its sensitivity to habitat fragmentation. **Recommended Next Step:** Conduct a Phase 1 habitat assessment focusing on vernal pool presence/connectivity, seasonal wetland conditions, and upland serpentine grassland, followed by focused species surveys during appropriate seasonal windows (typically spring for amphibians and shrimp). Results will determine whether incidental take permits, habitat conservation plans, or design modifications are needed to avoid triggering jeopardy conclusions. **For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.**

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

### NATIVE PLANTS - iNaturalist Research-Grade within 10mi

| 58 | 5052 | 2026-07-04 |
|---|---|---|
| Unique Species | Total Observations | Most Recent |

| COMMON NAME | *SCIENTIFIC NAME* | LAST OBS |
|---|---|---|
| American black nightshade | *Solanum americanum* | 2026-05-29 |
| Bearded Clover | *Trifolium barbigerum* | 2026-05-15 |
| bird's foot cliffbrake | *Pellaea mucronata* | 2026-04-28 |
| Blow Wives | *Achyrachaena mollis* | 2026-05-20 |
| blue oak | *Quercus douglasii* | 2026-05-09 |
| Bolander's Phacelia | *Phacelia bolanderi* | 2026-04-27 |
| box elder | *Acer negundo* | 2026-05-18 |
| Braun's Giant Horsetail | *Equisetum telmateia braunii* | 2026-06-22 |
| California bay | *Umbellularia californica* | 2026-06-24 |
| California beeplant | *Scrophularia californica* | 2026-05-20 |
| California black oak | *Quercus kelloggii* | 2026-05-15 |
| California buckeye | *Aesculus californica* | 2026-06-02 |
| California Dutchman's Pipe | *Aristolochia californica* | 2026-05-26 |
| California milkwort | *Rhinotropis californica* | 2026-05-15 |
| California mugwort | *Artemisia douglasiana* | 2026-06-02 |
| California poppy | *Eschscholzia californica* | 2026-06-24 |
| California Wild Rose | *Rosa californica* | 2026-05-09 |
| Canyon liveforever | *Dudleya cymosa* | 2026-05-30 |
| - | *Castilleja densiflora densiflora* | 2026-05-09 |
| clay mariposa lily | *Calochortus argillosus* | 2026-05-15 |
| coast live oak | *Quercus agrifolia* | 2026-05-13 |
| coastal woodfern | *Dryopteris arguta* | 2026-05-20 |
| common selfheal | *Prunella vulgaris* | 2026-06-22 |
| common yarrow | *Achillea millefolium* | 2026-05-18 |
| Diogenes' lantern | *Calochortus amabilis* | 2026-05-15 |
| distant phacelia | *Phacelia distans* | 2026-05-15 |
| - | *Downingia concolor concolor* | 2026-05-15 |
| Dwarf Brodiaea | *Brodiaea terrestris* | 2026-05-15 |
| Elegant Clarkia | *Clarkia unguiculata* | 2026-05-16 |
| goldback fern | *Pentagramma triangularis* | 2026-05-20 |
| Ithuriel's Spear | *Triteleia laxa* | 2026-06-15 |
| maroonspot calicoflower | *Downingia concolor* | 2026-05-15 |
| narrowleaf milkweed | *Asclepias fascicularis* | 2026-06-25 |
| narrowleaf mule-ears | *Wyethia angustifolia* | 2026-05-15 |
| Narrowleaf Willow | *Salix exigua* | 2026-06-24 |
| needleleaf navarretia | *Navarretia intertexta* | 2026-05-15 |
| northern California black walnut | *Juglans hindsii* | 2026-06-24 |
| orange bush monkeyflower | *Diplacus aurantiacus* | 2026-05-29 |
| Oregon Ash | *Fraxinus latifolia* | 2026-06-02 |
| Pacific Bleeding Heart | *Dicentra formosa* | 2026-05-04 |
| Pacific madrone | *Arbutus menziesii* | 2026-07-04 |
| Pacific poison oak | *Toxicodendron diversilobum* | 2026-06-02 |
| pineapple-weed | *Matricaria discoidea* | 2026-05-20 |
| Redwood Sorrel | *Oxalis smalliana* | 2026-05-28 |
| small-headed clover | *Trifolium microcephalum* | 2026-05-15 |
| Superb Mariposa Lily | *Calochortus superbus* | 2026-06-11 |
| thimble clover | *Trifolium microdon* | 2026-05-15 |
| tomcat clover | *Trifolium willdenovii* | 2026-05-15 |
| Toyon | *Heteromeles arbutifolia* | 2026-06-19 |
| trailing blackberry | *Rubus ursinus* | 2026-06-24 |
| Tree Clover | *Trifolium ciliolatum* | 2026-05-15 |

| COMMON NAME | *SCIENTIFIC NAME* | LAST OBS |
|---|---|---|
| valley oak | *Quercus lobata* | 2026-06-24 |
| wavy-leafed soap plant | *Chlorogalum pomeridianum* | 2026-05-20 |
| western sycamore | *Platanus racemosa* | 2026-05-17 |
| white brodiaea | *Triteleia hyacinthina* | 2026-05-15 |
| white-tipped clover | *Trifolium variegatum* | 2026-05-15 |
| Winecup Clarkia | *Clarkia purpurea* | 2026-05-15 |
| Yellow Mariposa Lily | *Calochortus luteus* | 2026-06-11 |

**AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU**

Ecological Assessment: 38.3459, -122.3047 This site supports a mixed oak woodland and chaparral community, dominated by blue oak and California black oak with understory shrubs like California buckeye and California bay, typical of the North Bay foothills. The diverse species composition—including moisture indicators like Braun's Giant Horsetail and clay mariposa lily alongside drought-tolerant species like canyon liveforever—suggests variable soil moisture across the site, with likely seasonal seepage areas or north-facing slopes retaining moisture longer than south-facing aspects. For grading and drainage planning, expect fine-grained soils with variable permeability; poorly-drained pockets may develop in concave topography while adjacent upslope areas drain rapidly, requiring careful cut-and-fill design to avoid altering existing drainage patterns that support the current plant community. Consult a licensed botanist for site-specific vegetation assessment.

Source: iNaturalist research-grade observations - Native species filter applied

## FIELD HAZARDS - Venomous & Toxic Species within 25mi

**637** Venomous Snakes

**156** Venomous Spiders

**91** Stinging Insects

**1439** Toxic Plants

### VENOMOUS SNAKES

| *SPECIES* | OBSERVATIONS |
|---|---|
| *Crotalus oreganus oreganus* | 637 obs |

### VENOMOUS SPIDERS

| *SPECIES* | OBSERVATIONS |
|---|---|
| *Latrodectus hesperus* | 156 obs |

### STINGING INSECTS

| *SPECIES* | OBSERVATIONS |
|---|---|
| *Dasymutilla aureola* | 91 obs |
| *Dasymutilla californica* | 91 obs |
| *Dasymutilla sackenii* | 91 obs |

### TOXIC PLANTS

| *SPECIES* | OBSERVATIONS |
|---|---|
| *Toxicodendron diversilobum* | 1100 obs |
| *Datura wrightii* | 57 obs |
| *Datura stramonium* | 57 obs |
| *Datura ferox* | 57 obs |
| *Conium maculatum* | 199 obs |
| *Urtica gracilis* | 83 obs |
| *Urtica gracilis holosericea* | 83 obs |

**AI FIELD SAFETY BRIEFING - CLAUDE HAIKU**

Field Safety Briefing – Napa County Area Your highest priorities are poison oak (extremely common at 1,439 observations) and Northern Pacific rattlesnakes (637 observations)—both pose significant crew risk. For poison oak, wear long sleeves and pants, learn to identify the three-leaflet pattern, and wash exposed skin immediately after work; for rattlesnakes, stay alert during warm months (April–October), watch where you step and place your hands, wear boots, and give snakes distance if encountered. Watch for toxic Datura plants with trumpet-shaped flowers and avoid touching them, and be cautious of black widow spiders in dark, undisturbed spaces like equipment boxes. Late spring through early fall brings peak snake activity and plant toxicity, so increase vigilance during these months and consider scheduling fieldwork for cooler morning hours when snakes are less active. Always verify current conditions locally before fieldwork.

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

## FCC MICROWAVE PATHS — Licensed beam paths crossing within 1320ft of site

**3** Beam Paths

**545ft** Closest Beam

**3** Fixed P2P

| TX CALL | RX CALL | TYPE | PATH MI | CLOSEST |
|---|---|---|---|---|
| **WBO61** | WBO59 | Fixed Point-to-Point | — | 545ft |

| TX CALL | RX CALL | TYPE | PATH MI | CLOSEST |
|---|---|---|---|---|
| **KYH37** | WQAT627 | Fixed Point-to-Point | — | 926ft |
| **WRCR329** | WQAT627 | Fixed Point-to-Point | — | 1289ft |

Source: FCC ULS microwave license database — PostGIS spatial query — Paths within 1320ft (0.25mi) of site — Towers may be 50+ miles away — FCC Part 101 licensed paths

### FAA AIRWAYS — ATS Routes crossing site area

**26** Total Airways

**13** Jet Routes

**2** Victor Airways

**11** Other Routes

| AIRWAY | TYPE | LEVEL | MEA EAST | MEA WEST | TRACK | REV TRACK |
|---|---|---|---|---|---|---|
| **J1** | Jet Route | High | 18,000 ft | — | 303.8° | 121.4° |
| **J110** | Jet Route | High | 18,000 ft | — | 137.9° | 318.2° |
| **J126** | Jet Route | High | 18,000 ft | — | 325.5° | 144.0° |
| **J143** | Jet Route | High | 18,000 ft | — | 325.0° | 145.8° |
| **J3** | Jet Route | High | 18,000 ft | — | 342.8° | 161.8° |
| **J32** | Jet Route | High | 18,000 ft | — | 19.2° | 199.6° |
| **J501** | Jet Route | High | 18,000 ft | — | 317.1° | 135.4° |
| **J58** | Jet Route | High | 18,000 ft | — | 65.3° | 245.9° |
| **J65** | Jet Route | High | 18,000 ft | — | 304.1° | 121.1° |
| **J80** | Jet Route | High | 18,000 ft | — | 65.3° | 245.9° |
| **J84** | Jet Route | High | 18,000 ft | — | 50.8° | 231.0° |
| **J88** | Jet Route | High | 18,000 ft | — | 308.0° | 127.3° |
| **J94** | Jet Route | High | 18,000 ft | — | 65.3° | 245.9° |
| **Q1** | Q-Route | High | — | — | 334.5° | 154.1° |
| **Q120** | Q-Route | High | — | — | 38.1° | 219.8° |
| **Q122** | Q-Route | High | — | — | 37.8° | 219.6° |
| **Q124** | Q-Route | High | — | — | 37.8° | 219.6° |
| **Q138** | Q-Route | High | — | — | 47.7° | 229.0° |
| **Q3** | Q-Route | High | — | — | 164.0° | 344.6° |
| **Q5** | Q-Route | High | — | — | 161.4° | 341.7° |
| **T257** | T-Route | Low | — | — | 321.1° | 140.5° |
| **T259** | T-Route | Low | — | — | 41.6° | 221.8° |
| **T261** | T-Route | Low | — | — | 31.4° | 211.7° |
| **T263** | T-Route | Low | — | — | 308.8° | 128.2° |
| **V107** | Victor Airway | Low | 7,000 ft | — | 295.4° | 114.2° |
| **V108** | Victor Airway | Low | 4,500 ft | — | 117.7° | 296.8° |

Source: FAA ATS Route FeatureServer — Airways shown cross the site bounding area and may not pass directly overhead — MEA = Minimum En Route Altitude

### RAILROAD PROXIMITY — FRA North American Rail Network

**8** Rail Lines

| OWNER / OPERATOR | TYPE | TRACKS | PASSENGER | STRACNET | SUBDIVISION |
|---|---|---|---|---|---|
| **CFNR** | Other | 1 | No | No | — |
| **CFNR** | Yard | 1 | No | No | — |
| **CFNR** | Mainline | 1 | No | No | VALLEJO |
| **SMRT** | Mainline | 1 | No | No | BRAZOS JCT. |
| **NVRR** | Mainline | 1 | No | No | MARTINEZ |
| **USG** | Other | 1 | No | No | — |
| **CFNR** | S | 1 | No | No | — |
| **NVRR** | Yard | 1 | No | No | — |

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

### WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

**200** Total Observations

**88** Total Species

**64** Birds

**6** Mammals

**12** Reptiles

**6** Amphibians

**BIRDS (64 species)**

| COMMON NAME | *Scientific Name* | OBS |
|---|---|---|
| **California Towhee** | *Melozone crissalis* | 7 |
| **California Quail** | *Callipepla californica* | 6 |
| **Wild Turkey** | *Meleagris gallopavo* | 6 |
| **Snowy Egret** | *Egretta thula* | 6 |
| **Western Bluebird** | *Sialia mexicana* | 5 |
| **Turkey Vulture** | *Cathartes aura* | 4 |
| **Ash-throated Flycatcher** | *Myiarchus cinerascens* | 4 |
| **Great Egret** | *Ardea alba* | 4 |
| **Red-tailed Hawk** | *Buteo jamaicensis* | 4 |
| **Black Phoebe** | *Sayornis nigricans* | 4 |
| **Red-shouldered Hawk** | *Buteo lineatus* | 3 |
| **Northern Flicker** | *Colaptes auratus* | 3 |
| **Cooper's Hawk** | *Astur cooperii* | 3 |
| **Western Screech-Owl** | *Megascops kennicottii* | 3 |
| **Mallard** | *Anas platyrhynchos* | 3 |
| *+ 49 additional species* | | |

**MAMMALS (6 species)**

| COMMON NAME | *Scientific Name* | OBS |
|---|---|---|
| **Coyote** | *Canis latrans* | 3 |
| **Black-tailed Jackrabbit** | *Lepus californicus* | 2 |
| **Mule Deer** | *Odocoileus hemionus* | 2 |
| **Domestic Cat** | *Felis catus* | 1 |
| **Eastern Fox Squirrel** | *Sciurus niger* | 1 |
| **Striped Skunk** | *Mephitis mephitis* | 1 |

**REPTILES (12 species)**

| COMMON NAME | *Scientific Name* | OBS |
|---|---|---|
| **Western Fence Lizard** | *Sceloporus occidentalis* | 11 |
| **Northern Pacific Rattlesnake** | *Crotalus oreganus oreganus* | 7 |
| **Pacific Gopher Snake** | *Pituophis catenifer catenifer* | 7 |
| **California King Snake** | *Lampropeltis californiae* | 3 |
| **Pond Slider** | *Trachemys scripta* | 2 |
| **Western Skink** | *Plestiodon skiltonianus* | 2 |
| **Ring-necked Snake** | *Diadophis punctatus* | 1 |
| **Common Garter Snake** | *Thamnophis sirtalis* | 1 |
| **Western Yellow-bellied Racer** | *Coluber constrictor mormon* | 1 |
| **Pacific Ringneck Snake** | *Diadophis punctatus amabilis* | 1 |
| **Northwestern Pond Turtle** | *Actinemys marmorata* | 1 |
| **Common Sharp-tailed Snake** | *Contia tenuis* | 1 |

**AMPHIBIANS (6 species)**

| COMMON NAME | *Scientific Name* | OBS |
|---|---|---|
| **Pacific chorus frog** | *Pseudacris regilla* | 4 |
| **Western Toad** | *Anaxyrus boreas* | 3 |
| **American Bullfrog** | *Lithobates catesbeianus* | 3 |
| **Foothill Yellow-legged Frog** | *Rana boylii* | 3 |
| **Rough-skinned Newt** | *Taricha granulosa* | 1 |
| **California Giant Salamander** | *Dicamptodon ensatus* | 1 |

**AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU**

Ecological Character Assessment The 25-mile radius centered at 38.3459, -122.3047 (north of San Francisco Bay) represents a mixed foothill and valley ecosystem supporting a diverse wildlife community of 88 species, including year-round residents and seasonal migrants typical of California's Mediterranean climate zone. Species of regulatory concern include the Foothill Yellow-legged Frog (CDFG Species of Special Concern), Rough-skinned Newt (moderate toxicity risk), and Northwestern Pond Turtle (declining populations), indicating the presence of sensitive aquatic and riparian habitats requiring protection. The amphibian and reptile diversity, particularly aquatic species (frogs, turtles, garter snakes), suggests proximity to seasonal or perennial water features with intact vegetative structure; the bird community (waterfowl, raptors, flycatchers) and mammalian diversity indicate a mosaic of grasslands, oak woodland, and shrubland with moderate connectivity and variable disturbance. The prevalence of native species alongside adaptable generalists (domestic cat, Eastern Fox Squirrel) and the absence of sensitive species typically found in undisturbed habitats suggests the site has experienced historical land use conversion but retains functional habitat connectivity and water availability important for regional biodiversity. **For planning purposes only. Consult a qualified biologist for site-specific biological assessments.**

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

## ☀ PASSIVE + ACTIVE SOLAR PLANNING — Lat 38.35°

### 1 · AVERAGE SUN HOURS

| Period | Peak Hrs/Day | Notes |
|---|---|---|
| Annual average | 6.6 hrs | Overall solar resource |
| Summer average | 9 hrs | June–August |
| Winter average | 3.9 hrs | December–February |
| Worst month (Dec) | 3.2 hrs | Size battery storage to this |

### 2 · IDEAL SOLAR ARRAY PITCH

Optimal tilt = **38.3°** (latitude) → **9/12 pitch**

| Pitch | Angle | % of Optimal | Notes |
|---|---|---|---|
| 4/12 | 18.4° | 97% | good |
| 5/12 | 22.6° | 97.6% | good |
| 6/12 | 26.6° | 98.2% | ← excellent |
| 7/12 | 30.3° | 98.8% | ← excellent |
| 8/12 | 33.7° | 99.3% | ← excellent |
| 9/12 | 36.9° | 99.8% | ← annual optimal |
| 10/12 | 39.8° | 99.8% | ← excellent |
| 12/12 | 45° | 99% | ← excellent |

Going from 5/12 to 6/12 at framing costs ~$800–1,200 one time. On a 10kW system that's +723 kWh/yr (~$108 value/yr). Over 25yr panel life: ~$2710 total gain.

### 3 · SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

| Surface | Winter | Summer | Design Note |
|---|---|---|---|
| South glass (SHGC 0.4) | 51 | 31 | Minimize south glass |
| South glass (SHGC 0.6) | 76 | 46 | Clear/uncoated glass |
| North glass (any) | 10 | 13 | Negligible — diffuse only |
| East / West glass | 27 | 104 | ⚠ Minimize — afternoon load |
| Roof — dark (absorb 0.95) | 236 | 301 | Highest load surface |
| Roof — light (absorb 0.35) | 87 | 111 | Cool roof — significant reduction |
| Roof — metal (absorb 0.25) | 62 | 79 | Best roof option for heat |
| South wall — dark | 94 | 69 | |
| South wall — light | 31 | 23 | Light color = major savings |
| Hardscape — asphalt | 181 | 301 | ⚠ Heat island — avoid near structure |
| Hardscape — concrete | 124 | 206 | Also reflects onto adjacent walls |
| Hardscape — light pave | 67 | 111 | Recommended within 50ft |
| Pool surface | 23 | 38 | Low absorb + evaporative cooling |

Concrete reflection onto south wall: ~4 BTU/hr·ft² additional load per 100sf of adjacent hardscape.
Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

### 4 · ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

| Pitch | Dark Roof | Light Roof | Reduction vs Flat |
|---|---|---|---|
| 0/12 | 301 BTU | 111 BTU | baseline (worst) |
| 4/12 | 241 BTU | 89 BTU | -20% dark / -20% light |
| 6/12 | 196 BTU | 72 BTU | -35% dark / -35% light |
| 8/12 | 166 BTU | 61 BTU | -45% dark / -45% light |
| 12/12 | 135 BTU | 50 BTU | -55% dark / -55% light |

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

### 5 · THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

| Assembly | Lag | Peak Interior Load | Note |
|---|---|---|---|
| 2x4 frame (R-15) | 0.6 hrs | ~15:36 PM | R-value only — no mass benefit |
| 2x6 frame (R-23) | 0.9 hrs | ~15:54 PM | R-value only — no mass benefit |
| 2x8 frame (R-30) | 1.2 hrs | ~16:12 PM | R-value only — no mass benefit |
| 2x10 frame (R-37) | 1.5 hrs | ~16:30 PM | good |
| 2x12 frame (R-44) | 1.8 hrs | ~16:48 PM | good |
| ICF 6" core | 7 hrs | ~22:00 PM | ✓ shifts past sunset — natural vent window |
| 8" CMU block | 6 hrs | ~21:00 PM | ✓ shifts past sunset — natural vent window |
| 12" concrete | 8.5 hrs | ~23:30 PM | ✓ shifts past sunset — natural vent window |

In 3C Marine: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

### 6 · TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

| Tree Ht | Shadow Reaches | Summer hrs/day | Equinox hrs/day | Winter hrs/day | WUI note |
|---|---|---|---|---|---|
| 10ft | All year | 1.1 | 1 | 1.2 | OK |
| 20ft | All year | 2.1 | 1.9 | 2.4 | OK |
| 30ft | All year | 3.1 | 2.9 | 3.7 | Zone 2 check |
| 40ft | All year | 4 | 3.8 | 5.2 | ⚠ verify |
| 50ft | All year | 4.8 | 4.6 | 7.3 | ⚠ verify |
| 60ft | All year | 5.6 | 5.5 | 9.3 | ✗ Zone 1 |

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.
Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.
⚠ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

## 7 · ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

| Standard | Unbalanced (1/150) | Balanced† (1/300) | Notes |
|---|---|---|---|
| Code minimum | 8 sq in | 4 sq in | Moisture protection only |
| Thermal comfort | 24 sq in | 12 sq in | Keeps attic within 20°F of outdoor |
| WUI mesh 1/16" | 60 sq in | 30 sq in | 2.5× compensation — mesh clogs |
| WUI intumescent | 24 sq in | 12 sq in | Self-closing — no compensation needed |

**Pitch factor:** 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45
**Color factor:** Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45
**Scale:** multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor
**†Balanced** = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)
**WUI conflict:** Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.
⚠ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

### ⬡ AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Solar & Energy Planning for Your Home Your location gets about 6.6 hours of useful sunlight per day on average, which is solid for solar panels—think of it like having a part-time job that's most productive in summer (9 hours) and lightest in winter (under 4 hours). Your roof's ideal angle is about 38 degrees, which happens to match a 9-to-12 pitch, so if your roof already sits at that angle, you're in luck. Now here's where most people miss the bigger picture: a dark roof absorbs heat like a black car in the sun—it's pulling in 301 BTU of heat per hour per square foot on a summer day, but if you switched to a light-colored roof, that drops to just 111 BTU, which makes a real difference in your cooling bills. The west side of your home is your summer enemy because that afternoon sun throws 104 BTU of heat through any windows facing that direction, so shading, awnings, or even light curtains there will pay for themselves. Here's the elegant part: a big deciduous tree planted on your south side gives you the best of both worlds—it blocks about 4 hours of intense summer heat when you need the shade most, but in winter when the leaves drop, it only blocks 5 hours of the gentler winter sun, which you actually want coming through your south windows to warm the house naturally. Thermal mass—materials like concrete, stone, or thick walls—acts like a heat battery that soaks up warmth during the day and releases it slowly at night, smoothing out temperature swings. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.

## 🧠 SITE INTELLIGENCE SUMMARY AI synthesis — Claude Sonnet

⚙ Technical 👤 Client

*Technical Brief — for architects, engineers, and surveyors*

### STRUCTURAL DESIGN DRIVERS

This Napa County site falls within IECC Zone 3C (Marine) with an Annual Freezing Index of 0°F-days, placing it in a mild coastal climate with negligible frost action. Wind and snow load data are unavailable, precluding direct comparison of lateral load cases. However, the marine classification and zero AFI indicate that wind—not snow accumulation or freeze-thaw cycling—will likely govern the structural envelope if present in local code supplements. The absence of reportable wind and snow parameters suggests the design team should consult current Napa County building code amendments and site-specific meteorological records to establish the controlling load case; seismic design per current California standards will be a parallel requirement regardless of wind/snow governess.

### GEOTECHNICAL FOUNDATION STRATEGY

The site is underlain by Coombs gravelly loam with 2 to 5 percent slopes, classified in NRCS Hydrologic Group C with well-drained characteristics. Group C soils typically have moderate infiltration rates and support shallow spread footing systems where bearing capacity is adequate. The well-drained classification favors good drainage performance and eliminates standing water concerns typical of Groups D or A/D blends. Frost depth data are not available; the zero AFI suggests minimal frost penetration risk, but the design team must verify minimum footing depth requirements against Napa County code, which may impose a nominal frost line regardless of climatic calculation. Subsurface exploration (Standard Penetration Test or Cone Penetration Test) will be essential to confirm bearing capacity and confirm soil texture detail, as the provided SSURGO classification omits texture specification.

### HAZARD PROFILE

No Quaternary faults are registered within the search radius of this location, reducing local fault rupture risk. Subsidence basin mapping data are absent, indicating either low subsidence risk or lack of data coverage; further verification against USGS subsidence mapping is warranted if deep groundwater withdrawal or hydrocarbon extraction is planned. Eight oil and gas wells exist within approximately 10 miles, but the nearest (Mobil Oil Corporation, 7.2 miles distant) is plugged and inactive, eliminating active subsurface pressure or extraction-related hazards. Wildfire data were not provided; however, Napa County's fire history warrants confirmation of defensible space requirements and material flammability standards if not already incorporated in the project brief.

### SITE CONTEXT AND CLIMATE ENVELOPE

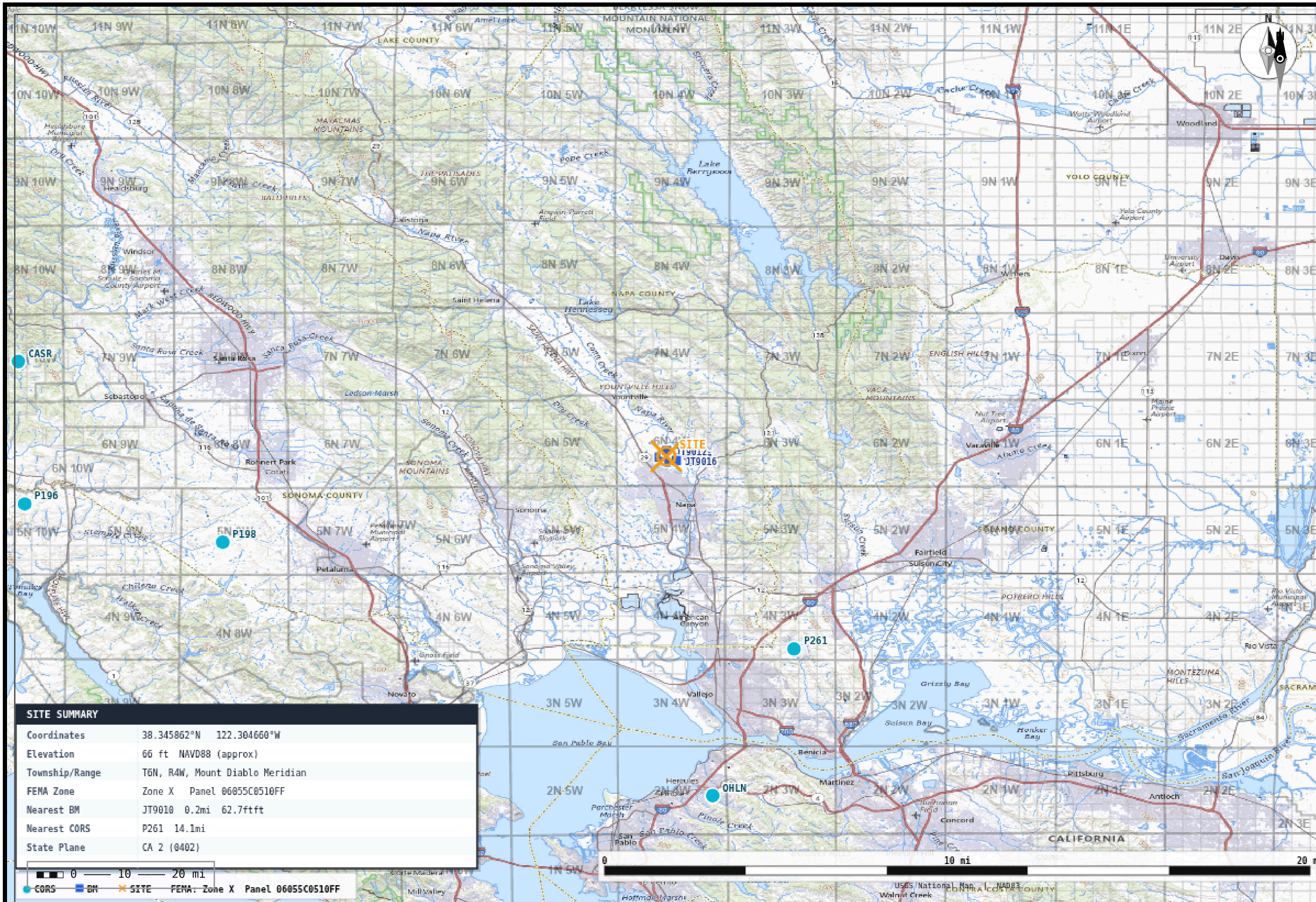
The marine climate zone (3C) with flat freezing index implies a narrow annual temperature swing and reduced exposure to thermal cycling stress. No mineral extraction records exist within the search radius, eliminating mine subsidence or legacy disturbance as design drivers. The 2 to 5 percent slope of the Coombs gravelly loam and well-drained classification support conventional site development with standard grading and stormwater management; slope stability analysis should be performed at detailed design if cuts exceed four feet. The marine climate envelope also sets a lower bar for thermal insulation compared to colder IECC zones, allowing focus on air leakage control and solar gain management per Zone 3C prescriptive requirements.

### NEXT STEP

The design team's first priority is obtaining site-specific wind and seismic design parameters from Napa County code and performing subsurface exploration (boring and SPT) to confirm bearing capacity and footing depth, since climatic and fault data do not constrain the structural system—soil capacity and code-mandated lateral loads will.

📋 Copy to clipboard

Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.




## SITE SUMMARY

| | |
|---|---|
| Coordinates | 38.345862°N 122.304660°W |
| Elevation | 66 ft NAVD88 (approx) |
| Township/Range | T6N, R4W, Mount Diablo Meridian |
| FEMA Zone | Zone X Panel 06055C0510FF |
| Nearest BM | JT9010 0.2mi 62.7ftft |
| Nearest CORS | P261 14.1mi |
| State Plane | CA 2 (0402) |

**TOPOSCOUT**
FIELD MAP

**TopoScout Field Map**
Lat: 38.34586 Lng: -122.30466
38.345862°N 122.304660°W | Radius: 30 mi

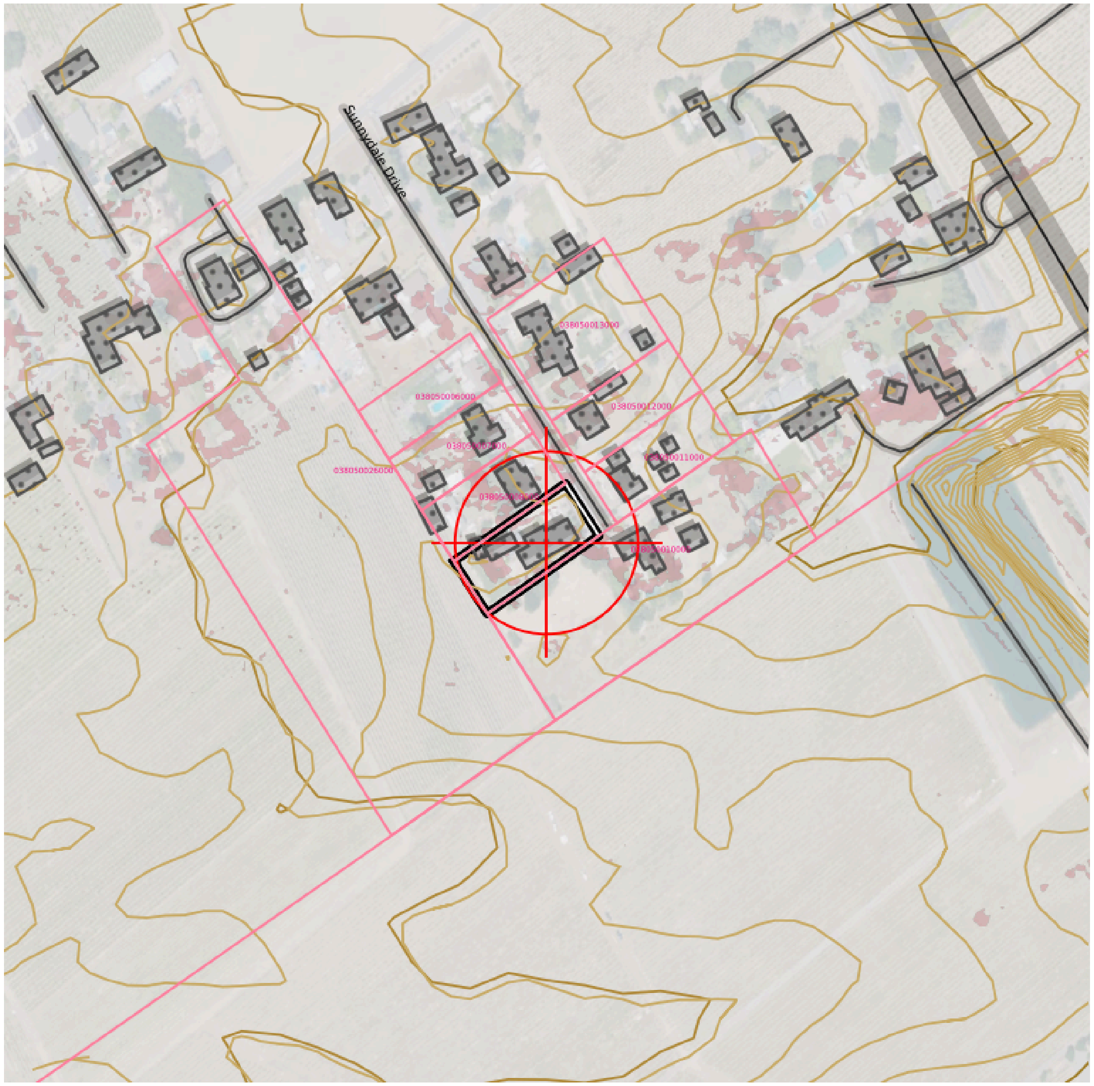
Generated: July 14, 2026
Base: USGS National Map
Generated by TopoScout · outKastdesigns.com

**TOPOSCOUT** SITE DXF PREVIEW

45 CAD Layers · AutoCAD Compatible · State Plane Coordinates

**Need the editable CAD file? Add the DXF at toposcout.com**

## TOPOSCOUT SITE PLAN PREVIEW

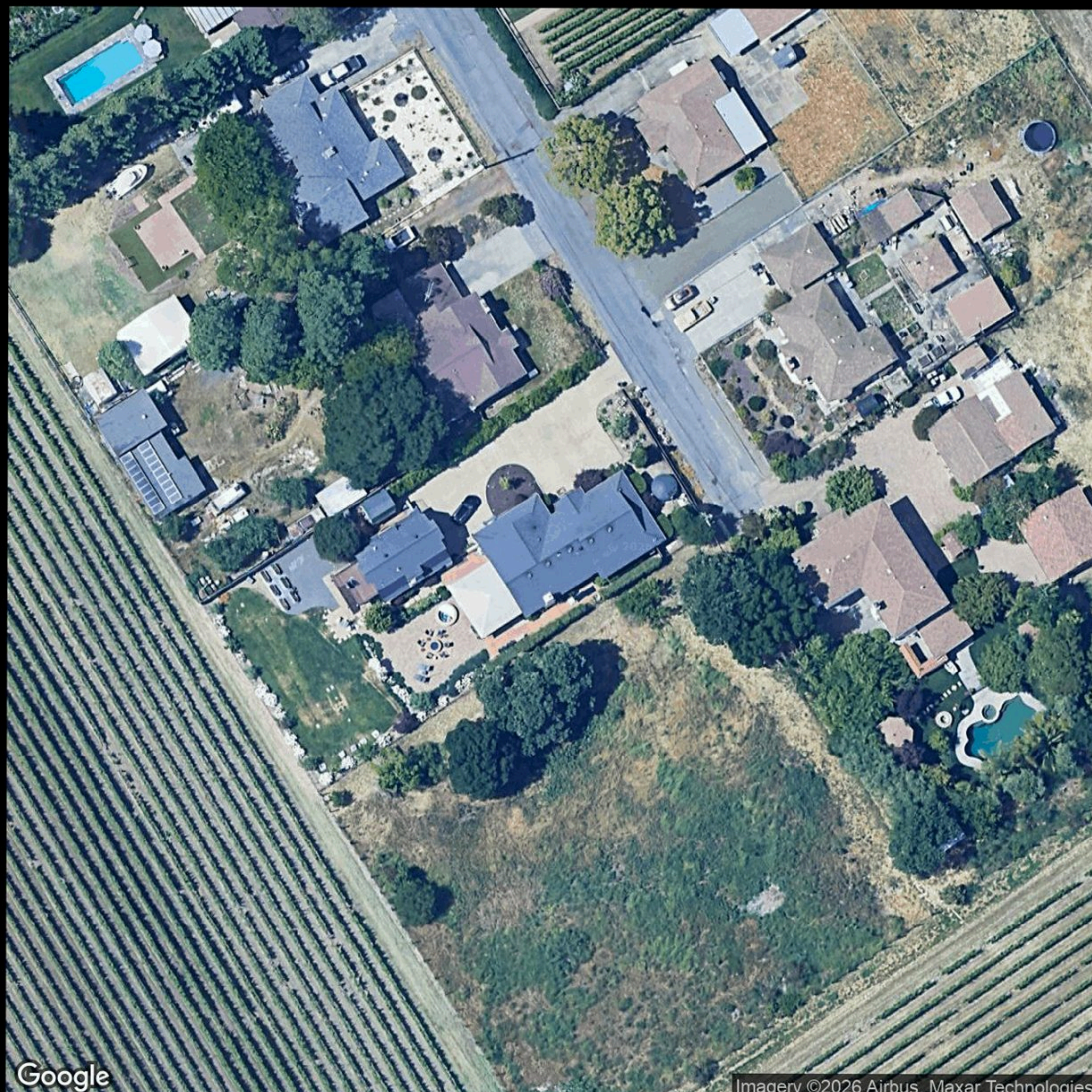



**INCLUDED LAYERS**

N

0 —— 400 ft State Plane · NAD83 · US Feet

■ Contours (Index/Major/Minor/1ft) ■ PLSS Section/Township/QQ ■ Roads Major/Local/Trail + ROW ■ Buildings ■ Hydro ■ Power Lines ■ Pipeline ■ Railroad ■ Benchmarks + Labels
■ County Boundary ■ OTLS Grid ■ Site Info ■ Elevation Labels ■ 3D Polylines (all geometry)




## SITE AERIAL — 600 ft × 600 ft

38.34586°N, 122.30466°W · Google Aerial

**TOPOSCOUT** SITE AERIAL IMAGERY

Source: Google Aerial

## AREA CONTEXT — 3000 ft × 3000 ft

38.34586°N, 122.30466°W · Google Aerial

## ⚡ FIELD RESOURCE LINKS

### PRE-MISSION PLANNING

**Trimble GNSS Planning Online** – https://trimblegnssplanningonline.com/
Satellite visibility, DOP charts, sky plots

**FAA B4UFLY** – https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly
Drone airspace authorization, TFRs, restricted areas

**Aloft (Kittyhawk)** – https://www.aloft.ai/
Field-friendly drone airspace map and LAANC authorization

**NOAA Space Weather — Live Kp** – https://www.swpc.noaa.gov/products/planetary-k-index
Real-time Kp index, geomagnetic storm alerts

### BASE STATION & NETWORK

**NOAA CORS Network** – https://geodesy.noaa.gov/CORS/cors-data.shtml
Continuously operating reference stations, data download

**NGS CORS Station Map** – https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml
Interactive map of all CORS stations with status

**IGS Station Monitor** – https://www.igs.org/network
International GNSS Service — global base station health

### POST-PROCESSING

**NGS OPUS** – https://geodesy.noaa.gov/OPUS/index.jsp
Online Positioning User Service — static GNSS processing

**NGS OPUS-RS** – https://geodesy.noaa.gov/OPUSI/index.jsp
Rapid static processing — sessions as short as 15 minutes

**CORS Data Download** – https://geodesy.noaa.gov/CORS/cors-data.shtml
Raw RINEX observation files from CORS stations

### REFERENCE & CALCULATIONS

**NGS Datasheets** – https://www.ngs.noaa.gov/cgi-bin/datasheet.prl
Benchmarks, control points, passive monuments

**NGS Coordinate Conversion (NCAT)** – https://geodesy.noaa.gov/NCAT/
NADCON5, datum transforms, state plane conversions

**NOAA Magnetic Declination** – https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml
Declination calculator for any location and date

**NGS Geoid Models** – https://geodesy.noaa.gov/GEOID/
GEOID18 and current model downloads, GEOID calculator

### FEMA & FLOOD

**FEMA Flood Map Service Center** – https://msc.fema.gov/portal/home
FIRM panels, flood zone lookup by address or coordinates

**FEMA Elevation Certificate** – https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate
Current EC form, instructions, LOMA/LOMR submission guidance

### DRONE OPERATIONS

**FAA Part 107 — Commercial Rules** – https://www.faa.gov/uas/commercial_operators
Full Part 107 regulatory summary for commercial UAS operations

**FAA DroneZone** – https://faadronezone.faa.gov/
Drone registration, operator certificates, waiver management

**FAA B4UFLY** – https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly
Pre-flight airspace check, TFRs, LAANC authorization

**FAA Part 107 Waivers** – https://www.faa.gov/uas/commercial_operators/part_107_waivers
Waiver applications for operations beyond standard 107 rules

**Aloft — Field Airspace Map** – https://www.aloft.ai/
Mobile-friendly real-time airspace, LAANC, flight logging

### AIR QUALITY & WILDFIRE SMOKE

AirNow Fire & Smoke Map
https://www.fire.airnow.gov
Current fire locations, smoke plumes, PM2.5 readings by location

AirNow Interactive Map
https://gispub.epa.gov/airnow
Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

NOAA Smoke Forecast
https://airquality.weather.gov
48-hr smoke concentration forecast, HRRR-Smoke model

Links verified: February 2026 · Check quarterly · TopoScout GNSS Planning System